

Classes n Object Assignments

1. Create a class named 'Student' with String variable 'name' , integer variable 'roll_no'., String variable 'phone_no' and String variable 'address' a. Assign the value of roll_no as '2' and that of name as "John" by creating an object of the class Student. b. Assign and print the roll number, phone number and address of two students having names "Sam" and "John" respectively by creating two objects of class 'Student'.

```
public class Student {
    String name;
    int roll_no;
    String phone_no;
    String address;

    public static void main(String[] args) {
        Student s1 = new Student();
        s1.roll_no = 2;
        s1.name = "John";

        System.out.println("Roll No : " + s1.roll_no);
        System.out.println("Name : " + s1.name);

        Student s2 = new Student();
        Student s3 = new Student();

        s2.name = "Sam";
        s2.roll_no = 1;
        s2.phone_no = "9876543210";
        s2.address = "Mumbai";

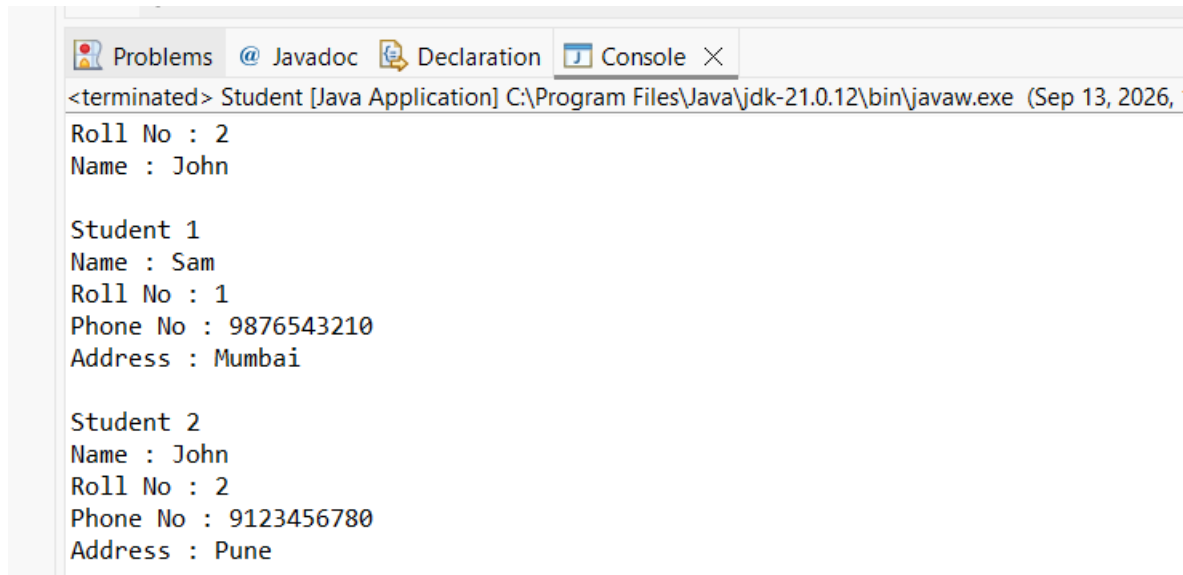
        s3.name = "John";
        s3.roll_no = 2;
        s3.phone_no = "9123456780";
        s3.address = "Pune";

        System.out.println("\nStudent 1");
        System.out.println("Name : " + s2.name);
        System.out.println("Roll No : " + s2.roll_no);
        System.out.println("Phone No : " + s2.phone_no);
        System.out.println("Address : " + s2.address);
    }
}
```

```

        System.out.println("\nStudent 2");
        System.out.println("Name : " + s3.name);
        System.out.println("Roll No : " + s3.roll_no);
        System.out.println("Phone No : " + s3.phone_no);
        System.out.println("Address : " + s3.address);
    }
}

```



```

<terminated> Student [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026,
Roll No : 2
Name : John

Student 1
Name : Sam
Roll No : 1
Phone No : 9876543210
Address : Mumbai

Student 2
Name : John
Roll No : 2
Phone No : 9123456780
Address : Pune

```

2. Write a program to print the area and perimeter of a triangle having sides of 3, 4 and 5 units by creating a class named 'Triangle' with constructor having the three sides as its parameters.

```

public class Triangle {
    int side1;
    int side2;
    int side3;
    public Triangle(int side1, int side2, int side3) {

        this.side1 = side1;
        this.side2 = side2;
        this.side3 = side3;
    }
    public int perimeter() {

        return side1 + side2 + side3;
    }
}

```

```

public double area() {

    double s = (side1 + side2 + side3) / 2.0;

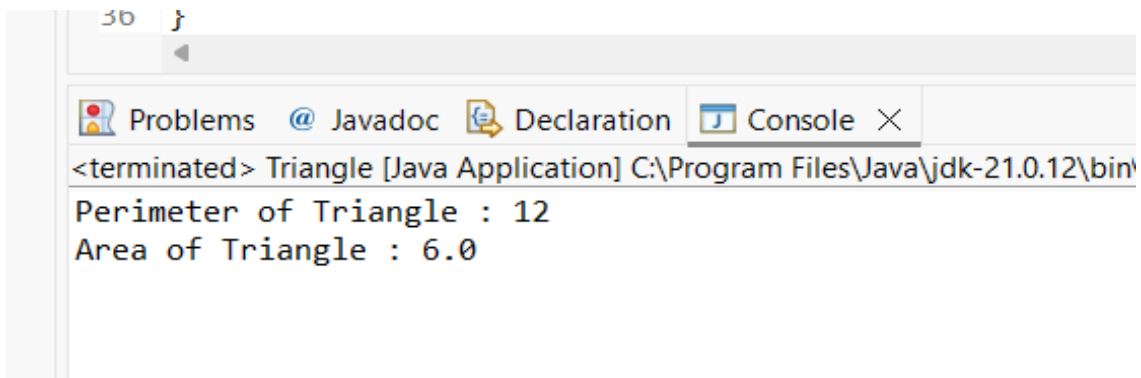
    return Math.sqrt(s * (s - side1) * (s - side2) * (s - side3));
}

public static void main(String[] args) {

    Triangle t = new Triangle(3, 4, 5);

    System.out.println("Perimeter of Triangle : " + t.perimeter());
    System.out.println("Area of Triangle : " + t.area());
}
}

```



3. Write a program to print the area of two rectangles having sides (4,5) and (5,8) respectively by creating a class named 'Rectangle' with a method named 'area' which returns the area and length and breadth passed as parameters to its constructor.

```

public class Rectangle {

    int length;
    int breadth;
    public Rectangle(int length, int breadth) {
        this.length = length;
        this.breadth = breadth;
    }
}

```

```

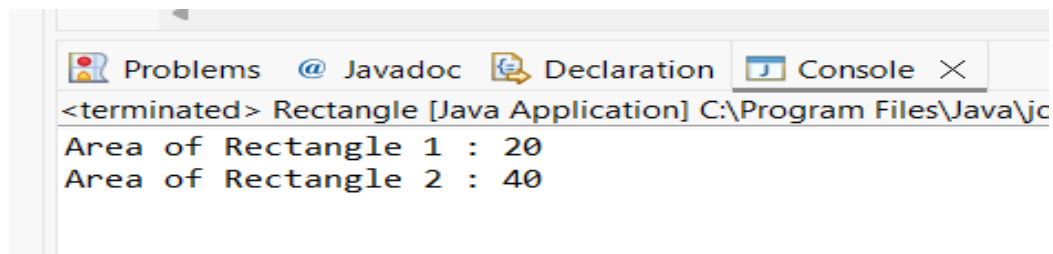
public int area() {
    return length * breadth;
}

public static void main(String[] args) {

    Rectangle r1 = new Rectangle(4, 5);
    Rectangle r2 = new Rectangle(5, 8);

    System.out.println("Area of Rectangle 1 : " + r1.area());
    System.out.println("Area of Rectangle 2 : " + r2.area());
}
}

```



4. **Print the sum, difference and product of two complex numbers by creating a class named 'Complex' with separate methods for each operation whose real and imaginary parts are entered by user.**

```

public class Complex {
    int real;
    int imaginary;
    public Complex(int real, int imaginary) {
        this.real = real;
        this.imaginary = imaginary;
    }
    public void sum(Complex c) {
        int r = real + c.real;
        int i = imaginary + c.imaginary;

        System.out.println("Sum = " + r + " + " + i + "i");
    }
    public void difference(Complex c) {
        int r = real - c.real;
        int i = imaginary - c.imaginary;
    }
}

```

```

        System.out.println("Difference = " + r + " + " + i + "i");
    }
    public void product(Complex c) {
        int r = (real * c.real) - (imaginary * c.imaginary);
        int i = (real * c.imaginary) + (imaginary * c.real);

        System.out.println("Product = " + r + " + " + i + "i");
    }
    public static void main(String[] args) {

        System.out.println("Enter real part of first number:");
        int real1 = ConsoleInput.getInt();

        System.out.println("Enter imaginary part of first number:");
        int imaginary1 = ConsoleInput.getInt();

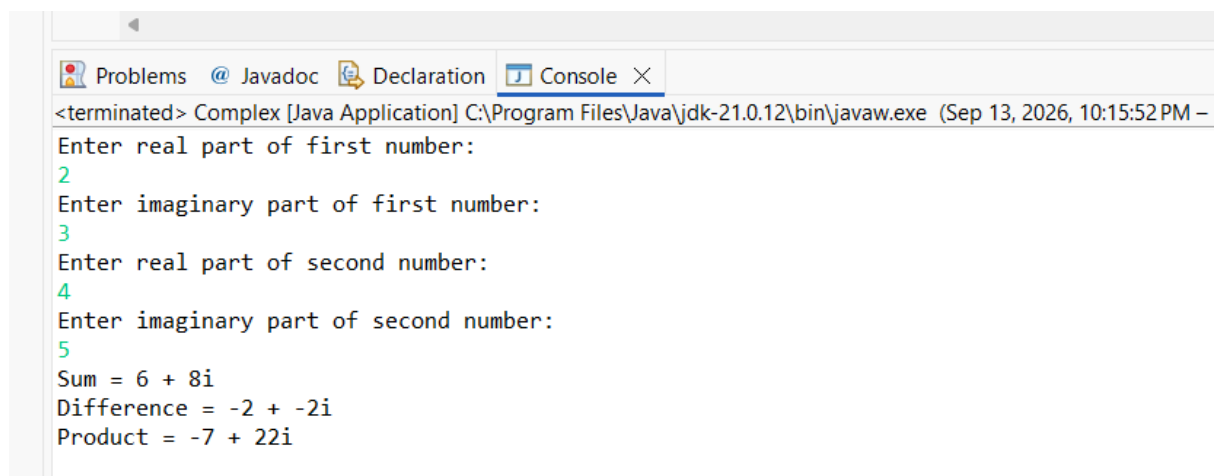
        System.out.println("Enter real part of second number:");
        int real2 = ConsoleInput.getInt();

        System.out.println("Enter imaginary part of second number:");
        int imaginary2 = ConsoleInput.getInt();

        Complex c1 = new Complex(real1, imaginary1);
        Complex c2 = new Complex(real2, imaginary2);

        c1.sum(c2);
        c1.difference(c2);
        c1.product(c2);
    }
}

```



```

<terminated> Complex [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026, 10:15:52 PM -
Enter real part of first number:
2
Enter imaginary part of first number:
3
Enter real part of second number:
4
Enter imaginary part of second number:
5
Sum = 6 + 8i
Difference = -2 + -2i
Product = -7 + 22i

```

5. Write a program that would print the information (name, year of joining, salary, address) of three employees by creating a class named 'Employee'. The output should be as follows:

Name	Year of joining	Address
Robert	1994	64C- WallsStreat
Sam	2000	68D- WallsStreat
John	1999	26B- WallsStreat

```
public class EmployeeDetails {

    String name;
    int yearOfJoining;
    double salary;
    String address;

    public EmployeeDetails(String name, int yearOfJoining, double salary,
String address) {

        this.name = name;
        this.yearOfJoining = yearOfJoining;
        this.salary = salary;
        this.address = address;
    }

    public void display() {

        System.out.println(name + "\t" + yearOfJoining + "\t\t" + salary + "\t\t" +
address);
    }

    public static void main(String[] args) {

        EmployeeDetails e1 = new EmployeeDetails("Robert", 1994, 50000,
"64C-WallStreet");

        EmployeeDetails e2 = new EmployeeDetails("Sam", 2000, 60000, "68D-
WallStreet");

        EmployeeDetails e3 = new EmployeeDetails("John", 1999, 55000, "26B-
WallStreet");

        System.out.println("Name\tYear of Joining\tSalary\t\tAddress");
```

```

        e1.display();
        e2.display();
        e3.display();
    }
}

```

Name	Year of Joining	Salary	Address
Robert	1994	50000.0	64C-WallStreet
Sam	2000	60000.0	68D-WallStreet
John	1999	55000.0	26B-WallStreet

6. Write a program by creating an 'Employee' class having the following methods and print the final salary. 1 - 'getInfo()' which takes the salary, number of hours of work per day of employee as parameter 2 - 'addSal()' which adds \$10 to salary of the employee if it is less than \$500. 3 - 'addWork()' which adds \$5 to salary of employee if the number of hours of work per day is more than 6 hours.

```

public class EmpInfo {
    double salary;
    int hours;
    public void getInfo(double salary, int hours) {
        this.salary = salary;
        this.hours = hours;
    }
    public void addSal() {
        if (salary < 500) {
            salary = salary + 10;
        }
    }

    public void addWork() {
        if (hours > 6) {
            salary = salary + 5;
        }
    }
}

```

```
public static void main(String[] args) {  
  
    EmpInfo e = new EmpInfo();  
  
    e.getInfo(400, 8);  
  
    e.addSal();  
    e.addWork();  
  
    System.out.println("Final Salary : $" + e.salary);  
}  
}
```

